Snowflake Project – Data Engineering & SQL Solution

Date	24 September 2026
Team ID	
Project Name	EV Market Analytics Dashboard
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	[Paste GitHub / GitLab / hosted link here]
Data Source(s) Link	EV dataset (CSV) covering vehicle sales, pricing, battery specifications, manufacturing country and customer ratings. Link: [add dataset link]
Database	EV_DB, with schemas RAW, STAGING, CURATED and MART
Warehouse	EV_WH (X-Small virtual warehouse, auto-suspend 60 seconds)
Tables / Views Created	Table: RAW.EV_DATA_RAW (loaded data), STAGING.EV_DATA_STG (cleaned data), CURATED.EV_DATA_CLEAN (final data with derived fields). Views:

	MART.VW_CATEGORY_SALES, MART.VW_MANUFACTURER_SHARE, MART.VW_BATTERY_SUMMARY
Key Transformations / Calculations	NULL handling with COALESCE and filters; duplicate removal with ROW_NUMBER() and QUALIFY; trimming and standardisation of manufacturer, category, country and battery type; removal of invalid values; derived fields RANGE_PER_KWH, ESTIMATED_REVENUE_USD and RATING_BAND
KPIs created	Total annual sales, average driving range, average battery capacity, average customer rating, market share %, year-over-year sales growth %, number of manufacturers; Streamlit dashboard with 3 pages (Executive Overview, Manufacturer & Market Share, Battery & Technology)
Key Insights	[Add key findings from your dashboard, e.g. the leading vehicle category, the manufacturer with the highest market share, the most adopted battery type, the country with the highest production]

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	Yes
2	Data is cleaned, transformed, and prepared appropriately	Yes
3	Required tables, views, and data relationships are correctly implemented	Yes
4	Required SQL logic, calculations, and KPIs are	Yes

	correctly implemented	
5	Data and solution outputs are validated against the project requirements	Yes
6	Solution is properly organized, documented, and functional	Yes

Additional Notes / Comments:

The scripts are run in this order: (1) setup and load script below, (2) Data Preparation activities 1 to 4 (NULL handling, duplicates, cleaning, transformation and MART views), (3) Business Question queries, (4) Streamlit dashboard. Column names in the scripts are assumed and should match the dataset used.

SQL Setup and Load Script

```
-- 1. Database, warehouse and schemas
CREATE DATABASE IF NOT EXISTS EV_DB;
CREATE WAREHOUSE IF NOT EXISTS EV_WH
  WITH WAREHOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE;
USE DATABASE EV_DB;
USE WAREHOUSE EV_WH;
CREATE SCHEMA IF NOT EXISTS EV_DB.RAW;
CREATE SCHEMA IF NOT EXISTS EV_DB.STAGING;
CREATE SCHEMA IF NOT EXISTS EV_DB.CURATED;
CREATE SCHEMA IF NOT EXISTS EV_DB.MART;

-- 2. File format and stage for the CSV upload
CREATE OR REPLACE FILE FORMAT EV_DB.RAW.EV_CSV_FORMAT
  TYPE = 'CSV' FIELD_OPTIONALLY_ENCLOSED_BY = '"' SKIP_HEADER = 1
  NULL_IF = ('', 'NULL', 'NA', 'N/A');
CREATE OR REPLACE STAGE EV_DB.RAW.EV_STAGE FILE_FORMAT = EV_DB.RAW.EV_CSV_FORMAT;

-- 3. RAW table (column names assumed - match them to your dataset)
CREATE OR REPLACE TABLE EV_DB.RAW.EV_DATA_RAW (
  VEHICLE_ID            NUMBER,
  MANUFACTURER          VARCHAR,
  MODEL                 VARCHAR,
  VEHICLE_CATEGORY      VARCHAR,
  BATTERY_TYPE          VARCHAR,
  BATTERY_CAPACITY_KWH  NUMBER(8,2),
  DRIVING_RANGE_KM      NUMBER(8,2),
  PRICE_USD             NUMBER(12,2),
  ANNUAL_SALES          NUMBER,
  SALES_YEAR            NUMBER(4,0),
  MANUFACTURING_COUNTRY VARCHAR,
  CUSTOMER_RATING       NUMBER(3,1)
);

-- 4. Load the file (after uploading it to the stage) and check the row count
COPY INTO EV_DB.RAW.EV_DATA_RAW FROM @EV_DB.RAW.EV_STAGE ON_ERROR = 'CONTINUE';
SELECT COUNT(*) AS RAW_ROWS FROM EV_DB.RAW.EV_DATA_RAW;
```